# Decision Memo

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| TO | Sarah, Retail Analytics Lead |
| FROM | GiftLane Recommendation Project Team |
| DATE | October 13, 2026 |
| SUBJECT | Selecting a practical method for product recommendations |

**Recommendation**.
Use Truncated SVD as the current baseline for GiftLane's Top-10 product recommendations. It works directly with our sparse customer-product matrix and performs almost identically to PCA at 50 components. Before deployment, prioritize testing binary purchase values, which produced the strongest observed recommendation results.

## Tested

We cleaned 1,067,371 raw transaction rows to 802,920 valid rows, reserved the last two months for testing, and built a 5,475-customer by 3,689-product training matrix. The matrix is 97.95% empty, so CSR storage uses 4.76 MiB instead of 154.09 MiB for a dense representation (96.91% less storage). This is a sparse-storage saving, not an SVD compression figure.

## Measured performance

| Configuration | Hit Rate@10 | Other result |
|---|---|---|
| Best-seller baseline | 17.41% | Reference recommendation |
| PCA, k=50, log values | 37.27% | Fit 0.1021 s; 3.524 MiB factors |
| Truncated SVD, k=50, log | 37.32% | Fit 0.1057 s; 3.496 MiB factors |
| Truncated SVD, k=50, binary | 40.15% | Best tested purchase representation |

Hit Rate@10 counts an evaluated customer as a hit when at least one of ten recommendations matches a held-out purchase (1,913 eligible customers). PCA and SVD have nearly identical accuracy; in this run PCA used less peak process memory (207.92 versus 234.64 MiB). Process memory includes libraries and other overhead, not just model fitting.

## Information lost and trade-offs

Reducing 3,689 product features to 50 latent dimensions discards details in the original purchase matrix. At k=50, SVD explained 34.88% of the measured variance and had 78.94% relative reconstruction error; those figures are not recommendation error rates. Increasing k preserved more information but did not consistently improve Precision@10. SVD avoids PCA's mean-centering step for sparse input, although the measured fit times were practically identical.

## Risks and next steps

Risks: New customers/products lack usable training histories; evaluation excludes unavailable items and therefore does not measure every future purchase; customer segments are unevenly sized, including one group of only 12 customers. Next steps: Recheck holdout integrity, compare binary values at more component counts, measure full nightly pipeline runtime and memory on production-like hardware, and pilot recommendations before release.

Source: UCI Online Retail II (Chen, 2012). Figures from the project's reproduced DE2-ML7 experiments; timing and process RSS are machine-dependent.